Churnika Marappa Reddy

Software Engineer | Data Engineering • Machine Learning • AI/LLM Applications • Python • SQL • Cloud

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Education

University at Buffalo, SUNY | *M.S. in Engineering Science (Data Science)* — Dec 2026

Amrita Vishwa Vidyapeetham | *B.Tech in Computer Science and Engineering* — May 2025

Skills

- **Languages:** Python, SQL, Java, TypeScript, JavaScript, Scala, C++
- **Data Engineering:** ETL/ELT Pipelines, Apache Spark (PySpark), Airflow, Kafka, Apache Iceberg, dbt, Databricks, Snowflake, Data Quality
- **Machine Learning & AI:** PyTorch, TensorFlow, scikit-learn, XGBoost, MLflow, LLMs, RAG, LangGraph, LangChain, Vector Search (Qdrant), NLP
- **Backend, Cloud & DevOps:** FastAPI, REST APIs, Microservices, PostgreSQL, Supabase, Docker, Kubernetes, AWS, Azure, CI/CD (GitHub Actions)
- **Engineering Practices:** Unit & Integration Testing (pytest, Playwright), Concurrency Control, Code Review, Debugging, Git, Agile/Scrum

Experience

AI Engineer Intern, Backend & ML Integration | Product Manager Accelerator (PMA Apply) — Aug 2026 – Present

- Built core backend services for an AI job application platform, from pasted job link to submitted application, using Python, FastAPI, PostgreSQL and Next.js with TypeScript.
- Designed an LLM-powered match-scoring microservice in FastAPI and Pydantic, with 2 swappable providers, live LLM and deterministic, so tests run without model calls.
- Developed a resume-parsing pipeline (pypdf, python-docx, regex NLP) that loads PDF and DOCX profile data into PostgreSQL behind Supabase JWT auth, and debugged 3 production parsing issues.
- Implemented a shared data model with section-scoped writes and optimistic concurrency control, letting 3 independent backend services safely update the same application record.
- Shipped a Manifest V3 Chrome extension that auto-fills Greenhouse application forms from the approved application, keeping the final submit decision with the student.
- Authored 280+ pytest unit and integration tests and led a 25+ file refactor with no regressions, agreeing shared API contracts with a multi-service team.

Research Assistant, Generative Handwriting & Document Recognition | University at Buffalo — Mar 2026 – May 2026

- Engineered data and training pipelines with custom loaders for 2 handwriting datasets, IAM and GNHK.
- Adapted a few-shot diffusion model in PyTorch and benchmarked 2 generative approaches, diffusion and Transformer, on unseen writers.
- Created a Streamlit inference app with style-based ranking that loads checkpoints on 2 runtimes, CPU and GPU.

Web Development Intern | Tirumala Tirupati Devasthanam — Jun 2024 – Aug 2024

- Troubleshot frontend, backend and API defects across enterprise web applications, tracing root causes with developers and verifying fixes before release.
- Ran UI, API and data-flow tests and pre-release regression checks, documenting defects with reproduction steps.

Python Intern | Slash Mark — Oct 2023 – Feb 2024

- Wrote reusable Python programs for automation, data processing and analysis in a Git workflow.

Projects

Citi Bike Data Pipeline & Demand Forecasting | *PySpark, Airflow, Iceberg, LightGBM, FastAPI* — GitHub

- Delivered ETL infrastructure that transforms monthly trip data with PySpark and SQL into a 3-layer (Bronze/Silver/Gold) Apache Iceberg lakehouse, partitioned by year, month and station.
- Enforced data quality with a Write-Audit-Publish flow that stages each run in an isolated Iceberg table and fails it on a drop-rate quality gate.
- Tuned a cost-sensitive LightGBM demand model with Optuna and served predictions through FastAPI and Streamlit in Docker, cutting operational penalty 50.9% against a naive baseline.
- Orchestrated 5 Airflow DAGs with idempotent, re-runnable ingestion for 4 workloads: batch loading, BI marts, ML datasets and scoring.

CareFlow AI, Agentic RAG Platform | *LangGraph, FastAPI, Next.js, PostgreSQL, Qdrant* — Live | GitHub

- Architected and deployed an agentic RAG app where a LangGraph agent routes questions over Medicare policy documents plus FHIR and claims data in PostgreSQL, reaching Hit@1 0.96.
- Optimized hybrid keyword and dense retrieval with rank fusion and a cross-encoder ranker, raising held-out MRR@5 from 0.78 to 0.83.
- Deployed on Vercel, Render, Neon and Qdrant Cloud, with 1,171 automated tests, 9 end-to-end journeys and a 6-job CI/CD pipeline gating every commit.
- Evaluated LLM outputs with citation-to-evidence validation and abstention, routing flagged answers to human review with an audit trail.

Telco Customer Churn Prediction | *scikit-learn, XGBoost, Optuna, MLflow, Docker* — GitHub

- Benchmarked 16 model configurations across 4 algorithms, Logistic Regression, Random Forest, SVC and XGBoost, with Optuna tuning, MLflow tracking and a Dockerized FastAPI service.